

VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science Engineering and Information Systems (SCORE)

FALL Semester 2026-2027

Software Configuration Management

Review 1

Project Title

AI-Based Medicine Reminder and Prescription Management System

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1. Problem Statement

Medication non-adherence is a widespread problem affecting patients across all age groups, particularly the elderly and individuals managing chronic illnesses such as diabetes, hypertension, and cardiac conditions. Missed doses, incorrect dosages, and expired prescriptions frequently lead to worsening health outcomes, avoidable hospital readmissions, and increased healthcare costs. Many patients also struggle to keep track of multiple prescriptions issued by different doctors, refill schedules, and interactions between medications.

Existing solutions are largely manual and fragmented — patients rely on paper prescriptions, memory, or simple phone alarms, none of which account for prescription validity, refill timing, or communication with caregivers and pharmacies. There is no centralized system that links doctors, patients, caregivers, and pharmacies, and enables real-time monitoring of medicine intake and prescription status.

The proposed AI-Based Medicine Reminder and Prescription Management System addresses these problems by using artificial intelligence to analyze prescription data, generate intelligent reminder schedules, detect potential drug interactions, and notify caregivers when doses are missed. The system allows doctors to issue digital prescriptions, patients to track their medication schedules, and pharmacies to manage refill requests — all through a centralized, secure platform. This improves medication adherence, reduces health risks, and enhances the overall quality of patient care.

2. Objectives

The objectives of the proposed system are:

1. Develop a centralized platform for managing prescriptions and medication schedules.
2. Use AI algorithms to generate personalized reminder schedules and detect potential drug interactions.
3. Connect patients, doctors, caregivers, and pharmacies in real time.
4. Automate medicine intake tracking, refill alerts, and prescription renewal notifications.
5. Generate analytical reports on medication adherence, missed doses, and health trends.

3. Expected Users

The intended users of the system include:

- Patients
- Doctors / Physicians
- Caregivers / Family Members
- Pharmacies
- Hospitals and Clinics
- System Administrator

4. Development Methodology

Selected Methodology: Agile Scrum Methodology

Justification

The Agile Scrum methodology is selected because the project requires continuous feature enhancement and regular feedback from different stakeholders, including patients, doctors, caregivers, and pharmacies. Agile enables the development team to deliver working software in short iterations, making it easier to add AI reminder models, drug interaction checks, notification systems, and user-requested improvements throughout the development process. Using Django as the backend framework complements this approach, since its modular "app" structure and built-in admin interface allow individual features (prescriptions, reminders, notifications) to be developed, tested, and integrated incrementally across sprints.

Scrum Phases

Sprint 1

- Requirement Gathering
- User Story Creation
- Product Backlog

Sprint 2

- UI/UX Design
- Database Design
- System Architecture

Sprint 3

- User Authentication Module
- Prescription Management Module
- Medicine Inventory Module

Sprint 4

- AI Reminder & Drug Interaction Engine
- Notification System
- Caregiver & Pharmacy Alert Module

Sprint 5

- Testing
- Bug Fixing
- Documentation

Sprint 6

- Deployment

- Maintenance
- Performance Monitoring

5. Configuration Identification

The following configuration items (CIs) will be identified and managed throughout the software development lifecycle.

Configuration Item	Description	Version
Project Proposal	Initial project approval document	v1.0
Software Requirement Specification (SRS)	Functional and non-functional requirements	v1.0
Software Configuration Management Plan	SCM planning document	v1.0
System Design Document	Architecture and UML diagrams	v1.0
Database Design	ER Diagram and SQL scripts	v1.0
Source Code	Backend and Frontend code	Git Version
AI Reminder & Interaction Module	Machine Learning model files	v1.0
User Interface Designs	Wireframes and UI prototypes	v1.0
Test Plan	Testing strategy	v1.0
Test Cases	Functional and integration test cases	v1.0
User Manual	End-user documentation	v1.0
Deployment Guide	Installation instructions	v1.0
Release Package	Software release files	v1.0

Version Control Strategy

The project will use Git as the version control system to maintain software integrity.

Repository Structure

```
MedicineReminderSystem/

├── docs/
│      ├── SRS.docx
│      ├── SCM.docx
│      ├── DesignDocument.docx
│
├── backend/                    (Django project)
│      ├── manage.py
│      ├── config/              (settings, urls, wsgi/asgi)
│      ├── apps/
│      │      ├── users/
│      │      ├── prescriptions/
│      │      ├── reminders/
│      │      ├── notifications/
│      │      └── pharmacy/
│      └── requirements.txt
│
├── frontend/                   (React.js)
│
├── database/                   (PostgreSQL schema & migrations)
│
├── AI_Model/                   (Python ML scripts, notebooks)
│
├── test/
│
├── release/
│
├── README.md
```

Branching Strategy

Branch	Purpose
main	Stable production code
develop	Integration branch
feature/*	New feature development
bugfix/*	Bug fixing
release/*	Release preparation
hotfix/*	Critical production fixes

Configuration Control Process

The project follows a structured change management process:

1. Submit Change Request.
2. Review by Project Manager.

3. Perform Impact Analysis.
4. Obtain Approval.
5. Update Source Code.
6. Execute Testing.
7. Update Documentation.
8. Commit changes to Git.
9. Create a New Software Version.
10. Deploy Updated Release.

Configuration Status Accounting

Configuration status accounting keeps track of every configuration item throughout the project.

The following information will be maintained:

- Version Number
- Configuration Item Name
- Date Modified
- Modified By
- Change Request Number
- Approval Status
- Current Release
- Repository Commit ID

Baselines

The project establishes the following baselines to maintain software consistency:

Baseline	Description
Requirements Baseline	Approved software requirements
Design Baseline	Finalized system architecture
Development Baseline	Stable implementation after coding
Testing Baseline	Verified and tested software
Release Baseline	Approved production version

Naming Conventions

Item	Naming Standard
Django App	prescriptions/ (lowercase, plural)
Models	Prescription, MedicineReminder (PascalCase, models.py)
Views / ViewSets	PrescriptionViewSet (views.py)
Serializers	PrescriptionSerializer (serializers.py)
Database	medicine_reminder_db
API Endpoints	/api/v1/prescriptions/, /api/v1/reminders/
Documents	SCM_v1.0.docx
Releases	MedRemind_v1.0
Git Tags	v1.0.0

Tools Used

Django (Python) is used as the primary backend framework due to its built-in admin panel, strong ORM, rapid development cycle, and seamless integration with Python-based AI/ML libraries such as Scikit-learn and TensorFlow — allowing the reminder and drug-interaction engine to run within the same technology ecosystem as the backend. Celery with Redis is used alongside Django to handle time-based background tasks such as sending medicine reminders and refill alerts at scheduled times.

Activity	Tool
Version Control	Git & GitHub
Project Management	Jira
Documentation	Microsoft Word
UI Design	Figma
Backend Framework	Django / Django REST Framework (Python)
Frontend	React.js
Database	PostgreSQL (Django ORM)
AI Prediction / Reminder Engine	Python (Scikit-learn, TensorFlow)
Task Scheduling / Background Jobs	Celery with Redis (for timed reminders)

Activity	Tool
Notification Services	Firebase Cloud Messaging / Twilio
Authentication	Django Allauth / Simple JWT
Testing	Django TestCase, Pytest-Django, Postman
Containerization	Docker
CI/CD	GitHub Actions

Conclusion

The AI-Based Medicine Reminder and Prescription Management System provides an innovative solution for improving medication adherence while supporting better patient health outcomes. By integrating artificial intelligence, real-time notifications, and centralized prescription management, the system enhances patient safety and reduces the risks associated with missed or incorrect medication intake. Applying Software Configuration Management (SCM) ensures that all project artifacts — including requirements, source code, design documents, test cases, and releases — are systematically identified, version-controlled, and maintained throughout the software development lifecycle, resulting in a reliable, high-quality software product.

This project is more distinctive than common academic topics such as library management, attendance systems, or hospital management, making it an excellent choice for an SCM report while demonstrating practical relevance to healthcare technology and modern software engineering practices.